Analyse IV - Summary

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1 Information

A concise and rigorous summary transcribing the most important concepts and theorems from the Analyse IV lecture at EPFL.

2 Holomorphic and Analytic Functions

2.1 Definitions

Let $\Omega \subseteq \mathbb{C}$ be an open set. A complex function $f : \Omega \to \mathbb{C}$ is said to be **complex differentiable** at a point $z_0 \in \Omega$ if the following limit exists:

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

If f is complex differentiable at *every* point of Ω, we say f is **holomorphic** on Ω.
If f is holomorphic on the entire complex plane ($\Omega = \mathbb{C}$), we call f an **entire function** (e.g., polynomials, e^z, $\sin(z)$).

3 The Cauchy-Riemann Equations

Let $z = x + iy \in \mathbb{C}$. A complex function $f(z)$ can be separated into its real and imaginary parts:

$$f(x + iy) = u(x, y) + iv(x, y)$$

where $u, v : \mathbb{R}^2 \to \mathbb{R}$.

3.1 Necessary and Sufficient Conditions

Theorem: Suppose $u(x, y)$ and $v(x, y)$ have continuous first-order partial derivatives (i.e., they are of class C^1) on an open set Ω. The function $f = u + iv$ is holomorphic on Ω if and only if it satisfies the **Cauchy-Riemann equations**:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

When these equations hold, the complex derivative can be computed as:

$$f'(z) = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} - i\frac{\partial u}{\partial y}$$

3.2 The Complex Chain Rule

The chain rule for complex functions works exactly like the chain rule in real calculus. This is a vital tool for proving that a composite function is holomorphic without having to check the Cauchy-Riemann equations directly.

The Formula:
Let $g : \Omega \to U$ be holomorphic on $\Omega \subseteq \mathbb{C}$, and let $f : U \to \mathbb{C}$ be holomorphic on U. Then the composite function $h(z) = (f \circ g)(z) = f(g(z))$ is holomorphic on Ω, and its derivative is:

$$h'(z) = f'(g(z)) \cdot g'(z)$$

3.2.1 How to Use the Chain Rule

To compute the derivative of a complex composition, follow these steps:

1. **Identify the components:** Break the function into an "outer" function $f(w)$ and an "inner" function $g(z)$.
2. **Differentiate the outer function:** Compute $f'(w)$ with respect to its variable w.
3. **Differentiate the inner function:** Compute $g'(z)$ with respect to z.
4. **Substitute and Multiply:** Multiply the results together, replacing the variable w in $f'(w)$ with the original inner function $g(z)$.

Example: Differentiating $h(z) = e^{z^2}$

- Let the outer function be $f(w) = e^w \implies f'(w) = e^w$.
- Let the inner function be $g(z) = z^2 \implies g'(z) = 2z$.
- Applying the formula:

$$h'(z) = e^{g(z)} \cdot (2z) = 2ze^{z^2}$$

3.2.2 Example: Logarithmic Composition

Consider $f(z) = \log(1 + z^2)$ using the principal branch of the logarithm.

1. Domain and Holomorphicity:
The principal branch $\log(w)$ is holomorphic for $w \in \mathbb{C} \setminus (-\infty, 0]$. Thus, $f(z)$ is holomorphic where $1 + z^2$ is not a non-positive real number. This occurs when $z \neq iy$ for $|y| \geq 1$. The largest open subset of holomorphicity is:

$$\Omega = \mathbb{C} \setminus \{iy : y \in \mathbb{R}, |y| \geq 1\}$$

2. Derivative via Chain Rule:
Let $w = g(z) = 1 + z^2$. Then $f(z) = \log(w)$.

$$\begin{aligned} f'(z) &= \frac{d}{dw}(\log w) \cdot \frac{d}{dz}(1 + z^2) \\ &= \frac{1}{1 + z^2} \cdot 2z = \frac{2z}{1 + z^2} \end{aligned}$$

Note on Holomorphicity:
If g is holomorphic on Ω and f is holomorphic on $g(\Omega)$, then $f \circ g$ is guaranteed to be holomorphic on Ω. This allows us to conclude that functions like $\sin(e^z)$ or $\cos(z^3)$ are **entire functions** because they are compositions of entire functions.

3.3 Harmonic Conjugates

Because holomorphic functions are infinitely differentiable, u and v have continuous second-order derivatives. By applying the Cauchy-Riemann equations and Schwarz's theorem (symmetry of second derivatives), we get:

$$\Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad \text{and} \quad \Delta v = \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$$

Therefore, the real and imaginary parts of a holomorphic function are **harmonic functions**. We say that v is the harmonic conjugate of u.

4 The Complex Logarithm

4.1 Principal Argument

For any $z \neq 0$, the principal argument $\text{Arg}(z)$ is the unique angle in the interval $(-\pi, \pi]$. It is computed using the arctangent function depending on the quadrant of (x, y).
Crucial limitation: $\text{Arg}(z)$ is discontinuous across the negative real axis.
this is how we compute it :

$$\text{Arg}(z) = \begin{cases} \arctan\left(\frac{y}{x}\right) & \text{if } x > 0 \\ \arctan\left(\frac{y}{x}\right) + \pi & \text{if } x < 0 \text{ and } y \geq 0 \\ \arctan\left(\frac{y}{x}\right) - \pi & \text{if } x < 0 \text{ and } y < 0 \\ \frac{\pi}{2} & \text{if } x = 0 \text{ and } y > 0 \\ -\frac{\pi}{2} & \text{if } x = 0 \text{ and } y < 0 \end{cases}$$

4.2 Principal Branch of the Logarithm

We define the **principal branch** of the complex logarithm, denoted $\text{Log}(z)$, as:

$$\text{Log}(z) = \ln|z| + i\text{Arg}(z) = \ln\left(\sqrt{x^2 + y^2}\right) + i\text{Arg}(x + iy)$$

Rigorous Properties:

- **Domain:** $\text{Log}(z)$ is a well-defined, single-valued function on the cut plane $\mathbb{C} \setminus (-\infty, 0]$. We must remove zero and the negative real axis (the branch cut) to maintain continuity.
- **Holomorphy:** On this cut plane, $\text{Log}(z)$ is holomorphic.
- **Derivative:** Its complex derivative is exactly what we expect from real analysis:

$$\frac{d}{dz}\text{Log}(z) = \frac{1}{z} \quad \text{for } z \in \mathbb{C} \setminus (-\infty, 0]$$

5 Lecture 3

6 Complex Integration

Definition (Line Integral)
Let $\gamma : [a, b] \to \mathbb{C}$ be a regular (smooth) curve that parametrizes the path $\Gamma \subseteq \mathbb{C}$. Let $f : \Gamma \to \mathbb{C}$ be a continuous function. We define the line integral of f along Γ by:

$$\int_\Gamma f(z)\, dz = \int_a^b f(\gamma(t))\gamma'(t)\, dt$$

Example
Let Γ be the unit semi-circle centered at $z = 0$, parametrized counterclockwise: $\gamma(t) = e^{it}$ for $t \in [0, \pi]$. Then $dz = ie^{it}dt$.

6.1 Cauchy's Theorem

Let $U \subseteq \mathbb{C}$ be an open and simply connected set, and let γ be a closed and piecewise regular curve inside U. If $f : U \to \mathbb{C}$ is holomorphic, then:

$$\oint_\Gamma f(z)\,dz = 0$$

- **Simply connected:** The set consists of "one piece" and has no holes.
- **Piecewise regular:** Γ consists of finitely many smooth curves joined end-to-end (e.g., the boundary of a square).

6.1.1 Example

Let Γ be the unit circle and $f(z) = z^2$. Since f is holomorphic on the simply connected set $U = \mathbb{C}$, by Cauchy's Theorem:

$$\oint_\Gamma z^2\,dz = 0$$

6.2 The Extension of Cauchy's Theorem (Deformation)

Let Γ_0, Γ_1 be closed, simple, regular curves such that $\Gamma_1 \subseteq \text{int}(\Gamma_0)$. Let γ_0, γ_1 be parametrizations of Γ_0, Γ_1 with the same orientation. If f is holomorphic on the region between Γ_0 and Γ_1 (including the boundaries), then:

$$\oint_{\Gamma_0} f(z)\,dz = \oint_{\Gamma_1} f(z)\,dz$$

6.3 Cauchy's Integral Formula

Suppose $D \subseteq \mathbb{C}$ is open and $f : D \to \mathbb{C}$ is holomorphic. If $z_0 \in D$, then:

$$f(z_0) = \frac{1}{2\pi i} \oint_\gamma \frac{f(z)}{z - z_0}\,dz$$

for every simple, closed, counterclockwise-oriented curve γ in D whose interior contains z_0.

7 Lecture 4

8 Tricks to Compute Integrals Easily

- **Cauchy's Theorem:** If f is holomorphic on a **simply connected** set D, then for any closed loop γ in D:

$$\int_\gamma f(z)\,dz = 0$$

- **Deformation Theorem:** We can replace an integral over a "complicated" curve Γ_0 by an integral over a "simpler" curve Γ_1 (like a small circle) as long as f is holomorphic in the region between Γ_0 and Γ_1.

- **Path Independence:** If Γ_0 and Γ_1 share the same endpoints and orientation, and f is holomorphic on the region they enclose, then:

$$\int_{\Gamma_0} f(z)\,dz = \int_{\Gamma_1} f(z)\,dz$$

- **Generalized Cauchy Integral Formula:** Useful for computing integrals of the form $\frac{f(z)}{(z-z_0)^{n+1}}$. If f is holomorphic on and inside γ:

$$\int_\gamma \frac{f(z)}{(z-z_0)^{n+1}}\,dz = \frac{2\pi i}{n!} f^{(n)}(z_0)$$

  Example: To calculate $\int_\gamma \frac{e^z}{z^3} dz$ around 0, use $n = 2$ and $f(z) = e^z$. The result is $\frac{2\pi i}{2!} f''(0) = \pi i$.

9 Taylor Series and Laurent Series

9.1 Taylor Series

Real case: Let $f : \mathbb{R} \to \mathbb{R}$ be $N+1$ times differentiable. For any $x_0 \in \mathbb{R}$, the Taylor polynomial is $T^N_{x_0} f(x)$. If $f \in \mathcal{C}^{N+1}$, there exists $\theta \in (0,1)$ such that:

$$f(x) = \sum_{n=0}^{N} \frac{f^{(n)}(x_0)}{n!}(x-x_0)^n + \frac{f^{(N+1)}(\theta x_0 + (1-\theta)x)}{(N+1)!}(x-x_0)^{N+1}$$

Complex case: Let $f : D \to \mathbb{C}$ be holomorphic. If the disk $B_R(z_0)$ is contained in D, then f is analytic and:

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!}(z-z_0)^n \quad \text{for all } z \in B_R(z_0)$$

The largest such R is the **Radius of Convergence**.

9.2 Laurent Series

Theorem: Let f be holomorphic on an annulus (punctured disk) $0 < |z-z_0| < R$. Then:

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z-z_0)^n = \cdots + \frac{c_{-2}}{(z-z_0)^2} + \frac{c_{-1}}{z-z_0} + c_0 + c_1(z-z_0) + \ldots$$

where the coefficients are given by:

$$c_n = \frac{1}{2\pi i} \int_\gamma \frac{f(w)}{(w-z_0)^{n+1}} dw$$

Definition: The series is split into two parts:

- **Singular part:** Terms with negative powers ($n < 0$).
- **Regular part:** Terms with non-negative powers ($n \geq 0$).

10 Classification of Singularities and Residues

10.1 Types of Isolated Singularities

By looking at the singular part of the Laurent series of f at z_0:

1. **Removable Singularity:** No negative powers ($c_n = 0$ for all $n < 0$).
2. **Pole of order k:** The lowest power is $c_{-k} \neq 0$ (finitely many negative terms).
3. **Essential Singularity:** Infinitely many negative terms (e.g., $e^{1/z}$ at $z = 0$).

11 Lecture 5

11.1 Definition

(1) We say f has **regular point** at z_0 if the singular part of the Laurent series at z_0 vanishes.
Example :

$$f(z) = \sum_{u=0}^{\infty} \frac{z^u}{u!} = e^z, \text{ 0 regular}$$

Remark: if f is holomorphic at z_0 then z_0 is a regular point. Then converte is also true: if z_0 is a regular point pf f, then $\lim_{z \to z_0} f(z)$ exists and if we extend f to z_0 this way, f is holomorphic at z_0. In short z_0 regular point of f if and only if $\lim_{z \to z_0} f(z)$ exists.

(2) we say f has a **pole of order** m at z_0 if the coefficient of the Laurent series of f at z_0 satisfy $c_n = 0$ for $u < -m$, ie

$$f(z) = \frac{c_{-m}}{(z - z_o)^m} + \frac{c_{-m+1}}{(z - z_0)^{m-1}} + \cdots, c_{-m} \neq 0$$

$$\text{"}f \text{ blows up with } \frac{1}{(z - z_0)^m} \text{ at } z_0\text{"}$$

Example

- $f(z) = \frac{1}{z}$ has a pole of order 1 at $z_0 = 0$
- $f(z) = \frac{1}{(z+2)^2} + \frac{1}{z}$ has a pole of order 2 at $z_0 = -2$ and a pole of order 1 at $z_0 = 0$

If f has a pole of order n at z_0,

$$f(z) = \frac{c_{-m}}{(z - z_0)^m} + \frac{c_{-m+1}}{(z - z_0)^{m-1}} + \cdots$$

Remarks $g(z) = (z - z_0)^m f(z)$ has a regular point at z_0 if f has a pole of order m at z_0.
In fact f has a pole of order m at z_0 if and only if $\lim_{z \to z_0}(z - z_0)^m f(z)$ exist and is not zero.
In particular if f has a pole of order z_0 we can write $f(z) = \frac{g(z)}{(z-z_0)^m}$ where g holomorphic at z_0 and $g(z_0) \neq 0$. (opposite is also true)

(3) we say f has an essential singularity at z_0 if its Laurent series has infinitely many coefficient $c_n \neq 0$ with $n \leq -1$
Example :

$$f(z) = e^{1/z} \text{ at } z_0 = 0.$$

$$f(z) = \sum_{u=0}^{\infty} \frac{(\frac{1}{z})^u}{u!} = \sum_{u=-\infty}^{0} \frac{1}{(-u)!} z^u$$

Example :

$$f(z) = ze^{1/z} \text{ at } z_0 = 0.$$

$$f(z) = z \sum_{u=-\infty}^{0} \frac{1}{(-u)!} z^u = z + 1 + \frac{1}{2!z} + \frac{1}{3!z} + \cdots$$

So 0 essentially singularity of f.
Now let's discuss quotients of Taylor series in this context. Suppose $p, q : D \setminus \{z_0\} \to \mathbb{C}$ holo with z_0 being regular for both. Write ther Laurent series as

$$p(z) = a_0 + a_1(z - z_0) + a_2(z - z_0)^2 + \cdots$$
$$q(z) = b_0 + b_1(z - z_0) + b_2(z - z_0)^2 + \cdots$$

Goal is to study (singularities of) the quotient $\frac{p}{q}$ near z_0. What is the order of the poles, if any? Let's discuss a few instructive examples.

1. if $b_0 \neq 0$ so $q(z_0) \neq 0$. In particular $\frac{1}{q}$ is holomorphic near z_0 and we have

$$\lim_{z - z_0} \frac{p(z)}{q(z)} = \frac{p(z_0)}{q(z_0)} = \frac{a_0}{b_0} \text{ exists, so } z_0 \text{ is a regular point for} \frac{p}{q}$$

2. if $a_0 = b_0 = 0$ but $b_1 \neq 0$ then

$$\lim_{z - z_0} \frac{p(z)}{q(z)} = \lim_{z - z_0} \frac{a_1(z - z_0) + a_2(z - z_0)^2 + \cdots}{b_1(z - z_0) + b_2(z - z_0)^2 + \cdots} = \lim_{z - z_0} \frac{a_1 + a_2(z - z_0)^2 + \cdots}{b_1 + b_2(z - z_0)^2 + \cdots} =$$
$$\frac{a_1}{b_1} \text{ exists hence } z_0 \text{ is a regular point for } f \text{ (l'Hôpital's rule)}$$

Similarly, if $a_0 = b_0 = 0$ and $a_1 = b_1 = 0$ yet $b_2 \neq 0$

Example :

$$f(z) = \frac{z}{sin(z)} \text{ where } z_0 = 0. \text{ using power series expansion of sin}$$

$$f(z) = \frac{z}{z - \frac{z^3}{3!} + \frac{z^5}{5!} + \cdots} = \frac{1}{1 - \frac{z^2}{3!} + \frac{z^4}{5!} + \cdots}$$

Hence $\lim_{z \to 0} f(z) = 1$ exist, so 0 regular point.
In summary either expand in power series and take out as many common powers of $z - z_0$ from the nominator and denominator as possible and take the limit or use l'Hôpital's rules.

12 Lecture 6, Residue theorem

12.1 Residues

Definition : The coefficient c_{-1} of the term $(z - z_0)^{-1}$ in the Laurent series for f at z_0 is called **residue of f at z_0**

$$Res_{z_0}(f) := c_{-1}$$

Remark : Note that by def of the coefficients in the Laurent series of f,

$$Res_{z_0}(f) := \frac{1}{2\pi i} \int_{\gamma} f(z)dz$$

where γ is a closed, simple, counterclockwise oriented curve that contains z_0 in its interior and f is holo on $int\gamma \setminus \{z_0\}$

12.2 Residue theorem

12.2.1 Theorem

Suppose $D \subseteq \mathbb{R}$ open and simply connected and let γ be simple closed curve in D which is counterclockwise oriented. Let $z_1, ..., z_w \in int\gamma$. Assume $f : D \backslash \{z_1, ..., z_w\} \to \mathbb{C}$ is holomorphic. Then

$$\int_\gamma f(z)dz = 2\pi i[Res_{z_1}(f) + ... + Res_{z_w}(f)]$$

12.2.2 Examples

Consider $f(z) = \frac{1}{z}$. Lez γ simple closed counterclockwise oriented curve in $\mathbb{C}$. Compute $\int_\gamma f dz$.

1. if $0 \in int\gamma$ then by residue thm.

$$\int_\gamma f dz = 2\pi i Res_0(f) = 2\pi i$$

2. if $0 \notin int\gamma$ then $\int_\gamma f dz = 0$
3. if $0 \in \gamma$ then $\int_\gamma f dz$ is not well-defined

12.3 Apllications

The residue thm can be used to compute integrals over simple closed curve, sometimes even to compute real integers by replacing thm with complex integrals!

12.3.1 Example

Compute $\int_{-\infty}^{\infty} \frac{1}{1+x^2} dx$. Set $f(z) = \frac{1}{1+z^2}$ Given $R > 0$ consider the segment $L_R : [-R, R] \to \mathbb{C}$ given by $L_R(t) = t$ and the half-circle $C_R : [0, \pi] \to \mathbb{C}$ given by $C_R(t) = Re^{it}$ Consider the closed curve $\gamma = L_R \cup C_R$. Then the function $f(z) = \frac{1}{1+z^2}$ has poles of order 1 at i. From the residue theorem:

$$\int_\gamma f dz = 2\pi i Res_i(f)$$

But

$$Res_i(f) = \lim_{z\to i}(z - i)f(z) = \lim_{z\to i} \frac{1}{z + i} = \frac{1}{2i}$$

13 Lecture 8, Fourier transform and Laplace transform

13.1 Review of Fourier analysis

The fourier transform well-defined is

$$\hat{f}(a) := \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-iax} f(x)dx$$

Sometimes $F[f]$ instead of $\hat{f}$

Inverse Fourier transform : $\overset{u}{f}$ well-defined is

$$\overset{u}{f}(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{iax} f(a)da$$

Sometimes $F^{-1}[f]$ instead.

13.1.1 Properties of Fourier transform

- Linearity : $F[\lambda f + \mu g] = \lambda F[f] + \mu F[g]$, whenever $\lambda, \mu \in \mathbb{R}$
- Time shift: Let $g(x) = f(x - x_0)$ then

$$\hat{g}(a) = e^{-iax_0} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-iay} f(y) dy = e^{-iax_0} \hat{f}(a)$$

- Frequency shift : Let $g(x) = e^{ia_0 x} f(x)$ for $a_0 \in \mathbb{R}$

$$\hat{g}(a) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(a-a_0)x} f(x) dx = \hat{f}(a - a_0)$$

- rescaling/dilation : Let $g(x) = f(\lambda x)$ where $\lambda \in \mathbb{R} \setminus \{0\}$

$$\hat{g}(a) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-iax} f(\lambda x) dx$$

  to recover $\hat{f}$ want to substitute $y = \lambda x$. Two cases

  - if $\lambda > 0$, we got $\int_{-\infty}^{\infty} \cdots \frac{1}{\lambda} dy$
  - if $\lambda < 0$, we got $\int_{\infty}^{-\infty} \cdots \frac{1}{\lambda} dy = -\int_{-\infty}^{\infty} \cdots \frac{1}{\lambda} dy = \int_{-\infty}^{\infty} \cdots \frac{1}{|\lambda|} dy$

  In both cases :

$$\hat{g}(a) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-iay/\lambda} f(y) \frac{1}{|\lambda|} dy = \frac{1}{|\lambda|} \hat{f}(\frac{a}{\lambda})$$

- derivatives : Let $g(x) = f'(x)$. Then

$$\hat{g}(a) = ia\hat{f}(a)$$

  More generally

$$F[f^{(u)}](a) = (ia)^u F[f](a)$$

  So then Fourier transform converts differential equation with constant coefficients into algebric equations on frequency domain.

13.1.2 Convolution

Given $f, g : \mathbb{R} \to \mathbb{C}$, the convolution $f * g$, whenever well-defined is

$$f * g(x) = \int_{-\infty}^{\infty} f(y) g(x - y) dy$$

Some properties of convolution:

- $f * g = g * f$
- $f * (g * h) = (f * g) * h$
- $f * (g + h) = f * g + f * h$
- $(f * g)' = f' * g = f * g'$

- Convolution :

$$F[f * g] = \sqrt{2\pi}F[f]F[g]$$

- Gaussian function : if $f(x) = e^{-x^2/2}$,

$$\hat{f}(a) = e^{-a^2/2}$$

- Plancherel's theorem :

$$\int_{-\infty}^{\infty} |f(x)|^2 dx = \int_{-\infty}^{\infty} |\hat{f}(a)|^2 da$$

13.2 Laplace analysis

Fourier transform computed for signals on $\mathbb{R}$
Laplace transform: Computed for signals defined on $\mathbb{R}_+$.
Both transform differential equ. into algebric equ.
Let $f : \mathbb{R}_+ \to \mathbb{C}$, where well-defined, the **Laplace transform** of f at $z \in \mathbb{C}$ is

$$\mathcal{L}[f](z) := \int_0^{\infty} f(t)e^{-zt}dt$$

Sometimes Laplace transform of functions are written by capital letters.

13.2.1 Convergence criteria

Of course the above integral well-edfined if $\int_{-\infty}^{\infty} |f(t)||e^{-zt}dt| < \infty$ Assume that is $\gamma_0 \in \mathbb{R}$ st

$$\int_0^{\infty} |f(t)|e^{-\gamma_0 t}dt < \infty$$

(This means $|f(t)| < e^{\gamma_0 t}$ as $t \to \infty$)

$$\begin{aligned} |\mathcal{L}[f](z)| &\leq \int_0^{\infty} |f(t)||e^{-zt}|dt \\ &= \int_0^{\infty} |f(t)||e^{-Re(z)t}|dt \\ &\leq \int_0^{\infty} |f(t)|e^{-\gamma_0 t}dt < \infty \end{aligned}$$

Thus if such a γ_0 exists, $\mathcal{L}[f](z)$ is defined at least for $z \in \{Re > \gamma_0\}$

13.2.2 Definition

A $\gamma_0 \in \mathbb{R}$ such that $\mathcal{L}[f](z)$ is well-defined at least on $\{Re > \gamma_0\}$ is called **abscission of convergence**
Let γ_0 abscisse of convergence. Thus $\mathcal{L}[f]$ is holomorphic on $\{Re > \gamma_0\}$. Set $F(z) = \mathcal{L}[f](z)$. Thus

$$\frac{d}{dz}F(z) = \frac{d}{dz}\int_0^{\infty} f(z)e^{-zt}dt$$
$$= \int_0^{\infty} f(t)(-t)e^{-zt}dt = -\mathcal{L}[tf(t)](z)$$

In short

$$\mathcal{L}[f(t)]' = -\mathcal{L}[tf(t)]$$

14 Lecture 9, some theorem

14.1 Theorem

For all $u \in \mathbb{N}$ and whenever $Re(z) > 0$

$$\mathcal{L}[t^u](z) = \frac{u!}{z^{u+1}}$$

14.1.1 Linearity

Let $f, g : \mathbb{R}_+ \to \mathbb{C}$ be st $\mathcal{L}[f], \mathcal{L}[g]$ are defined where $Re(z) > \gamma_0$ for some $\gamma_0 \in \mathbb{R}$. Then for all $\lambda, \mu \in \mathbb{C}$, $\mathcal{L}[\lambda f + \mu g](g)$ is defined where $Re(z) > \gamma_0$ and

$$\mathcal{L}[\lambda f + \mu g](z) = \lambda \mathcal{L}[f](z) + \mu \mathcal{L}[g](z)$$

14.2 Theorem

Let $f : \mathbb{R}_+ \to \mathbb{C}$ be of class C^1. Assume $\gamma_0 \in \mathbb{R}$ abscisse of coy for f and f'. Then for $Re(z) > \gamma_0$,

$$\mathcal{L}[f'](z) = z\mathcal{L}[f](z) - f(0)$$

Laplace trafo of derivation :

$$\mathcal{L}[f^{(u)}](z) = z^k \mathcal{L}[f](z) - \sum_{j=0}^{k-1} z^{k-1-j} f^{(j)}(0)$$

14.3 Theorem

Let $f : \mathbb{R}_+ \to \mathbb{C}$ piecewise continuous and $\gamma_0 \geq 0$ an abscissa of convergence for f. Define $g : \mathbb{R}_+ \to \mathbb{C}$ by $g(t) = \int_0^t f(s)ds$. Then when $Re(z) > \gamma_0$,

$$\mathcal{L}[g](z) = \frac{1}{z}\mathcal{L}[f](z)$$

14.4 Theorem

Let $a > 0$ and $b \in \mathbb{R}$. Let $f : \mathbb{R}_+ \to \mathbb{C}$ piecewise continuous with abscissa of coy $\gamma_0 \in \mathbb{R}$. Define $g : \mathbb{R}_+ \to \mathbb{C}$ by $g(t) = e^{-bt}f(at)$. Then

$$\mathcal{L}[g](z) = \frac{1}{a}\mathcal{L}[f](\frac{z+b}{a})$$

holds when $Re(\frac{z+b}{a}) > \gamma_0$

14.5 Theorem

Let $f, g : \mathbb{R}_+ \to \mathbb{C}$ piecewise continuous with common absissa of convergence $\gamma_0 \in \mathbb{R}$. THen γ_0 is an abscissa of convergence for $f \star g$ and where $Re(z) > \gamma_0$

$$\mathcal{L}[f \star g](z) = \mathcal{L}[f](z)\mathcal{L}[g](z)$$

15 Lecture 10, Application of Laplace theorem

15.1 Application of Laplace analysis to the Cauchy problem

15.1.1 ODE with constant coefficients

Let $a, y_0 \in \mathbb{R}$ and $f : \mathbb{R}_+ \to \mathbb{R}$ piecewise continuous. Task: Find a solution $y : \mathbb{R}_+ \to \mathbb{R}$ solving

$$(1) \begin{cases} y'(t) + ay(t) = f(t) \text{ for t ¿ 0} \\ y(0) = y_0 \text{ (initial condition)} \end{cases}$$

Existence: finding a solution (Laplace analysis

Uniqueness : there is a unique solution

Uniqueness for (1) Let $y_1, y_2 : \mathbb{R}_+ \to \mathbb{R}$ two sol to (1)

Show they coincide, we'll do this by showing $w(t) = 0$ for all $t \in \mathbb{R}_+$, where $w(t) = y_1(t) - y_2(t)$

$$\text{for } y_1, yields$$
$$\begin{cases} w'(t) + aw(t) = 0 \\ w(0) = 0 \end{cases}$$

with some multiplication, we can obtain

$$(e^{at}w(t))' = ae^{at}w(t) + e^{at}w'(t)$$

Hence $e^{at}w(t)$ const in t. Hence

$$e^{at}w(t) = e^{a0}w(0) = 0 \Rightarrow w(t) = 0$$

Existence: Find a solution by taking Laplace transformation of (1)

$$y'(t) + ay(t) = f(t)$$
$$\Rightarrow \mathcal{L}[y'](z) + a\mathcal{L}[y](z) = \mathcal{L}[f](z) \qquad (2)$$

Setting $Y = \mathcal{L}[y]$ and $F = \mathcal{L}[f]$ and using the formula for Laplce transformation of derivations :

$$(2) \Rightarrow zY(z) - y(0) + aY(z) = F(z)$$
$$\Rightarrow Y(z) = \frac{F(z)}{z+a} + \frac{y_0}{z+a}$$

With some changement of the function we can finally obtain

$$y(t) = \int_0^t e^{-a(t-s)} f(s)ds + y_0 e^{-at}$$

Second order initial value problem Let $w \in \mathbb{R}_+, a > 0$, and $f : \mathbb{R}_+ \to \mathbb{R}$ as above

$$(3) \begin{cases} y''(t) + 2ay'(t) + w^2 y(t) = f(t) \\ y(0) = y_0 \\ y'(0) = v_0 \end{cases}$$

The function is uniqueness

Existence We have three differenet cases :

1. $a^2 = w^2$ (critical damped) In this case :

$$y(t) = \int_0^t (t-s)e^{-a(t-s)} f(s)ds + e^{-at}y_0 + te^{-at}(ay_0 + v_0)$$

2. $w^2 > a^2$ (underdamped) we have

$$\begin{aligned}y(t) = \int_0^t \frac{1}{\sqrt{w^2-a^2}} e^{-a(t-s)} sin(\sqrt{w^2-a^2}(t-s))f(s)ds \\ +y_0 e^{-at} cos(\sqrt{w^2-a^2}t) \\ +(ay_0+v_0)\frac{1}{\sqrt{w^2-a^2}} e^{-at} sin(\sqrt{w^2-a^2}t)\end{aligned}$$

3. $w^2 < a^2$ (overdamped) Thus

$$\begin{aligned}y(t) = \int_0^t \frac{1}{\sqrt{a^2-w^2}} e^{-a(t-s)} sinh(\sqrt{a^2-w^2}(t-s))f(s)ds \\ +y_0 e^{-at} cosh(\sqrt{a^2-w^2}t) \\ +(ay_0+v_0)\frac{1}{\sqrt{a^2-w^2}} e^{-at} sinh(\sqrt{a^2-w^2}t)\end{aligned}$$

15.1.2 Inverse Laplace transformation

Theorem: Assume $f : \mathbb{R}_+ \to \mathbb{C}$ piecewise continuous and for some $\gamma \in \mathbb{R}$, $\mathcal{L}[f]$ is holo on $\{Re \geq \gamma\}$ and

$$\lim_{t\to\infty} \int_{-T}^{T} F(\gamma+is)e^{(\gamma+is)t} ds$$

exists for all $t \in \mathbb{R}_+$ Then

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \mathcal{L}[f](\gamma+is)e^{(\gamma+is)t} ds$$

16 Lecture 11,...

16.1 Example

$$F(z) = \frac{1}{(z+1)(z+2)^2}$$

Want to find $f : \mathbb{R}_+ \to \mathbb{C}$ auch that $\mathcal{L}[f] = F$ Two ways to do this:

1. Partial fonction decomposition

$$\begin{aligned}\frac{1}{(z+1)(z+2)^2} &= \frac{1}{z+1} - \frac{1}{z+2} - \frac{1}{(z+2)^2} \\ &= e^{-t} - e^{-2t} - te^{-2t}\end{aligned}$$

   Here is a few example how to do :

S.No.	Form of the rational function	Form of the partial fraction
1.	$\frac{px+q}{(x-a)(x-b)}, a \neq b$	$\frac{A}{x-a}+\frac{B}{x-b}$
2.	$\frac{px+q}{(x-a)^2}$	$\frac{A}{x-a}+\frac{B}{(x-a)^2}$
3.	$\frac{px^2+qx+r}{(x-a)(x-b)(x-c)}$	$\frac{A}{x-a}+\frac{B}{x-b}+\frac{C}{x-c}$
4.	$\frac{px^2+qx+r}{(x-a)^2(x-b)}$	$\frac{A}{x-a}+\frac{B}{(x-a)^2}+\frac{C}{x-b}$
5.	$\frac{px^2+qx+r}{(x-a)(x^2+bx+c)}$	$\frac{A}{x-a}+\frac{Bx+C}{x^2+bx+c}$
	• where $x^2 + bx +$ c cannot be factorised further	

2. Inverse laplace transdorm and residue theorem

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(\gamma+is)e^{(\gamma+is)t}ds$$

let $h(z) = F(z)e^{zt}$. Poles for h are $z = -1$ and $z = -2$ Now we choose $\gamma = 0$.

- F is holomorphic on $\{Re > -1\}$
- $\int_{-\infty}^{\infty} |F(\gamma + is)|ds < \infty$

Inverse Laplace transform

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} h(\gamma+is)ds$$

16.2 Applications to partial differential equations (PDES)

As for ODES we will apply the theory of Laplace and Fourier transform to solve initial-boundary value problems for **linear** PDES while **constant coefficients**,eg. licat equation

$$\begin{aligned}\frac{\partial u}{\partial t}(x,t) - \frac{\partial^2 u}{\partial x^2}u(x,t) = f(x,t) \quad &\text{for } t>0 \text{ and } x \in (0,L)\\ u(x,0) = u_0(x) \ &\text{(initial condition)}\\ u(0,t) = 0, u(L,t) = 0 \ &\text{(boundary condition)}\end{aligned}$$

Describe the evolution of temperature u of a one-dimensional rod of length L with heat source f and fixed temperature at the endpoints $0, L$

General procedure that we will follows

- **Fourier series** if space or time domain bounded
- **Fourier transform** if space or time domain $\mathbb{R}$
- **Laplace transform** if space or time domain $\mathbb{R}_+$

16.2.1 Fourier series recapitulation

We want to decompose $f : [0,1] \to \mathbb{R}$ as an infinite sum of ascillativy functions. Basics functions

- $$\sin(\frac{2\pi u}{L}x) \quad \text{where } u \geq 1$$

- $$\cos(\frac{2\pi u}{L}x) \text{ where } u \geq 0$$

Orthogonality Inner product :

$$< f, y >:= \int_0^L f(x)g(x)dx$$

- $$< \sin(\frac{2\pi u}{L}x), \sin(\frac{2\pi w}{L}x) >= \begin{cases} L/2 & \text{if w=u} \\ 0 & \text{otherwise} \end{cases}$$

- $$\text{some for } < \cos(\frac{2\pi u}{L}x), \cos(\frac{2\pi w}{L}x) >$$

- $$< \sin(\frac{2\pi u}{L}x), \cos(\frac{2\pi w}{L}x) >= 0 \text{ for all } u, w$$

We will use these fcts as orthogonal basis to decompose other functions via trigonometric (Fourier) series Given $f : [0, L] \to \mathbb{R}$ piecewise continuous, can be write

$$f(x) = \frac{a_0}{2} + \sum_{u=1}^{\infty}[a_u \cos(\frac{2\pi u}{L}x) + b_u \sin(\frac{2\pi u}{L}x)]$$

The answer is yes

To "guess" the form of the coefficients a_u and b_u, we assume f already has this form. Then we can (formally) compute the following using orthogonality.

- $$\int_0^L f(x)dx = \frac{L}{2}a_0$$

- $$\int_0^L f(x)\cos(\frac{2\pi u}{L}x)dx = \frac{L}{2}a_u \text{ where } u \geq 1$$

- $$\int_0^L f(x)\sin(\frac{2\pi u}{L}x)dx = \frac{L}{2}b_u \text{ where } u \geq 1$$

This suppests to define

$$a_u = \frac{2}{L}\int_0^L f(x)\cos(\frac{2\pi u}{L}x)dx \text{ where } u \geq 0$$
$$b_u = \frac{2}{L}\int_0^L f(x)\sin(\frac{2\pi u}{L}x)dx \text{ where } u \geq 1$$

Trigonomertic Fouries decomposition. Define $f_N : [0, L] \to \mathbb{R}$

$$f_N(x) = \frac{a_0}{2} + \sum_{u=1}^{N}[a_u \cos(\frac{2\pi u}{L}x) + b_u \sin(\frac{2\pi u}{L}x)dx]$$

where a_u and b_u are as above, ehre $f : [0, L] \to \mathbb{R}$ is piecewise continuous. Then

- $f_N(x) \to f(x)$ as $N \to \infty$ at every $x \in [0, L]$ where f is differentiable
- $f_N(x) \to \frac{f(x+)+f(x-)}{2}$ when x is a jump discoutinuity of f
- $$\lim_{u\to\infty} \int_0^L |f_N(x) - f(x)| dx = 0$$

16.2.2 special cases

- Assume f odd ($f(-x) = -f(x)$ for all $x \in [0, L]$) Then $a_u = 0$ for every $u \geq 0$ and

  $$f(x) = \sum_{u=1}^{\infty} b_u \sin(\frac{2\pi u}{L} x)$$

  where

  $$b_u = \frac{4}{L} \int_0^{\frac{L}{2}} f(x) \sin(\frac{2\pi u}{L} x) dx$$

- Assume f even. then $b_u = 0$ for all $u \geq 1$ and

  $$f(x) = \frac{a_0}{2} + \sum_{u=1}^{\infty} a_u \cos(\frac{2\pi u}{L} x)$$

  where

  $$a_u = \frac{4}{L} \int_0^{\frac{L}{2}} f(x) \cos(\frac{2\pi u}{L} x) dx$$